ProjectLedger

Academic Project Development & Continuous Monitoring Platform

Team 15 — Capstone Platform
VIT Vellore · Academic Year 2026
Live URL: https://projectledger-xg0l.onrender.com

Document Purpose: Complete operational handbook for student team members, faculty supervisors, and system administrators. Includes pre-seeded test accounts, feature workflows, and an end-to-end testing verification matrix.

1. Reference Seed Accounts & Credentials

The platform is pre-loaded with realistic academic accounts across all three roles. All default seed accounts share the same password.

Role	Full Name	Email Address	Password	Primary Access / Scope
Admin	Dr. Aris Thorne (Dean)	`admin@projectledger.edu`	`Admin@123`	Governance, faculty provisioning, global deadlines
Faculty	Dr. Ramesh Sharma	`faculty1@projectledger.edu`	`Faculty@123`	Guide for Project 1 & 3; reviews & rubrics
Faculty	Dr. Anita Roy	`faculty2@projectledger.edu`	`Faculty@123`	Guide for Project 2; blockchain reviews
Student (Lead)	Bawane Yash Sunil	`student1@projectledger.edu`	`Student@123`	Leader Project 1; tasks, milestones, GitHub sync
Student (Member)	Gulam Moinuddin	`student2@projectledger.edu`	`Student@123`	Member Project 1 & 3; task completion
Student (Member)	Kartik Jindal	`student3@projectledger.edu`	`Student@123`	Member Project 1; telemetry frontend tasks
Student (Lead)	Priya Nair	`student4@projectledger.edu`	`Student@123`	Leader Project 2 (At-Risk demo)
Student (Lead)	Siddharth Rao	`student6@projectledger.edu`	`Student@123`	Leader Project 3 (Delayed demo)

2. User Manual — Features & Workflows by Role

2.1 Student Portal Workflows

1. Registration & Login: Students self-register at /register using their University Email, Registration Number (e.g. 24BBS0119), and GitHub username. Once logged in, students land on their personalized Project Dashboard.

2. Creating a Project (Leader): Click *New Project* on the dashboard. Specify the Title, Problem Statement / Description, Dates, and select an approved Faculty Supervisor. The creator automatically becomes the **Project Leader**.

3. Team Roster Management: Leaders invite classmates to the team roster by their University Email or Registration Number and designate responsibilities (e.g., Backend Architect, ML Engineer, UI Designer).

4. Milestones Syllabus: Create structured academic deliverables (e.g. SRS Submission, Design Review, Prototype Demo). Each milestone has a target date and automatically aggregates linked tasks.

5. Task Board & Status Transitions: Tasks are assigned to teammates with priorities (LOW, MEDIUM, HIGH) and due dates. Team members change status from *NOT_STARTED* → *IN_PROGRESS* → *COMPLETED*. Progress updates automatically.

6. GitHub Telemetry Integration: Enter repository coordinates (e.g. repo_owner: spring-projects, repo_name: spring-petclinic). Click **Sync GitHub** to fetch commits and attribute contributions directly to teammates by their linked GitHub usernames.

2.2 Faculty Supervisor Workflows

1. Continuous Health Monitoring: Faculty dashboard displays all assigned capstone groups with dynamic health indicators: **ON TRACK** (progressing on schedule), **AT RISK** (milestone due within 7 days <50% done), and **DELAYED** (overdue milestones or high-priority tasks).

2. Milestone Review & Grading Rubrics: Faculty inspect deliverables, commit history, and task completion. Faculty approve milestones (marking them *REVIEWED*), leave feedback comments, and assign numeric scores (1.0 to 10.0).

2.3 Institutional Administrator Workflows

1. Faculty Provisioning: Faculty accounts cannot self-register. Administrators create and manage faculty accounts via the *Users* management console.

2. Institutional Deadlines: Admins post university-wide milestone freezes which immediately broadcast to all student and faculty dashboards.

3. Team End-to-End Testing Guide & Test Cases

Perform these test cases in sequence to verify all features and business logic across frontend, backend, and database persistence.

ID	Feature Area	Test Action / Scenario	Expected Verification Result
TC-01	**Authentication & Security**	Visit /login. Select 'Student' tab and sign in as student1@projectledger.edu. Attempt visiting /admin/dashboard directly.	Access to /admin/dashboard is blocked with HTTP 403 / Access Denied. RoleFilter enforces role boundaries.
TC-02	**Self-Registration Flow**	Click 'Create a student account' on /login. Fill all fields with a new email and valid password. Submit.	New student record is created in TiDB, password hashed with BCrypt, session created, and user lands on Student Dashboard.
TC-03	**Create Capstone Project**	As student leader, click '+ New Project'. Enter Title, Description, select Dr. Ramesh Sharma as Guide, and save.	Project is created with ON_TRACK status. Leader is assigned. Project details view renders milestone and team tabs.
TC-04	**Team Roster & Roles**	Inside project view, open 'Team' tab. Add teammate using student2@projectledger.edu with responsibility 'ML Engineer'.	Teammate is added to team_members table. Teammate can now see the project on their dashboard.
TC-05	**Milestone Creation**	In 'Milestones' tab, click 'Add Milestone'. Title: 'Architecture Design', Target Date: 2 weeks ahead. Save.	Milestone displays in syllabus list with PENDING review status and 0% progress.
TC-06	**Task Assignment & Progress**	Add 2 tasks under the milestone: Task A (HIGH priority), Task B (MEDIUM). Mark Task A COMPLETED.	Project progress percentage updates dynamically according to weighted formula. HealthCalculator recalculates status.
TC-07	**GitHub 1-Click Sync**	In 'Settings/Sync', enter Owner: 'octocat', Repo: 'Hello-World'. Click 'Sync GitHub Activity'.	Application contacts GitHub REST API v3, downloads commits, parses author logins, and logs records in github_activity table.
TC-08	**Faculty Review & Scoring**	Sign out and log in as faculty1@projectledger.edu. Open the project. Enter Review: 'Good architecture', Score: 9.0. Submit.	Review is saved in feedback table. Milestone status flips to REVIEWED. Score badge appears on student project view.
TC-09	**Admin User Provisioning**	Log in as admin@projectledger.edu. Navigate to 'Users' tab. Add new Faculty member 'Dr. Test Guide'.	New Faculty account is stored in database. Faculty can immediately sign in at /login with Faculty tab.
TC-10	**Global Deadline Broadcast**	As Admin, navigate to 'Deadlines'. Create a University Deadline 'Final Project Report' due in 30 days.	Deadline record is created without project_id constraint. Appears in upcoming deadlines list on all student dashboards.

4. Technical FAQ & System Architecture

Q1: What happens if Render puts the free service to sleep?

On the Render free tier, instances spin down after 15 minutes of inactivity. The first subsequent request triggers a cold-start of ~30-40 seconds. Setting up a free UptimeRobot (10-minute interval) pinging https://projectledger-xg0l.onrender.com/login keeps the instance warm 24/7.

Q2: How is Project Health calculated dynamically?

ProjectLedger utilizes an automated deterministic formula (HealthCalculator.java): If any HIGH-priority task is overdue by >3 days or an uncompleted milestone is past its target date, health becomes **DELAYED**. If a milestone is due within 7 days and task completion is <50%, health becomes **AT RISK**. Otherwise, it is **ON TRACK**.

Q3: Why does GitHub sync fail on private repositories?

The GitHub integration uses GitHub's public REST API v3. To sync private institutional repositories, configure a Personal Access Token in the service environment variable GITHUB_TOKEN=ghp_..., which also raises the API rate limit from 60 to 5,000 requests/hour.

Q4: Can student team members delete or rename projects?

Only the designated **Project Leader** or an **Administrator** has permission to edit project metadata or archive projects. General team members have permission to create/update tasks and view evaluations.

Q5: How does password security work?

All passwords are encrypted using salted BCrypt hashing (BCrypt.hashpw(plain, BCrypt.gensalt(12))). Cleartext passwords are never logged or stored in the database.

5. Technical Architecture Summary

- **Runtime & Backend:** Java 17 LTS, Jakarta Servlet 6.0, Jakarta Server Pages (JSP) 3.1, JSTL 3.0
- **Server Container:** Apache Tomcat 10.1 (Jakarta EE 10 namespace) deployed via multi-stage Dockerfile
- **Database Engine:** TiDB Cloud Serverless (MySQL 8.0 wire-compatible) with HikariCP connection pooling
- **Security Layer:** Per-session CSRF token validation, role-based URL filters, session-fixation protection
- **Frontend:** Handcrafted responsive CSS design system (Forest Green palette, Inter font, IBM Plex Mono)